

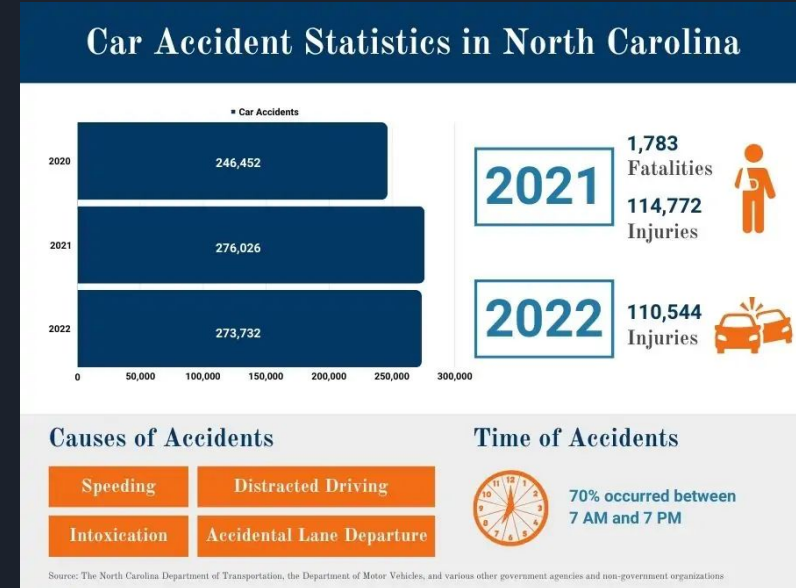


DTSC 2302 Final Project

Peyton Robinson, Brandon Miller, Emmett Lashbrook,
Esteban Grijalva, Yager Range

Introduction

From all the potential datasets available we decided on two we found as interesting. A dataset recording various car related incidents and a dataset recording population change in the city of Charlotte. We were curious as to how these two sets were potentially related, whether immigration and increase in population caused a linear increase in the crashes or maybe it was a more exponential increase. For regression, we wanted to model population change vs the number of traffic accidents. On the classification side we figured we could take this data and use it to find out high versus low traffic fatality risk years in Mecklenburg County. In our process we made several models for both regression and classification.



Model 1

This regression analysis was built to answer: how strongly do population change and recent crash history help explain annual traffic accidents in Mecklenburg County, the Charlotte proxy used for interpretation in the "County Estimate" data. To make the model strong, it was built not to rely on a single predictor or a single fitting method.

Linear Regression achieved the lowest error with an RMSE of 9,897.65 crashes, outperforming the nonlinear methods (Random Forest: 10,192, KNN: 10,225, GAM splines). Despite having higher cross validation error, Linear Regression generalized best to the held out test set.

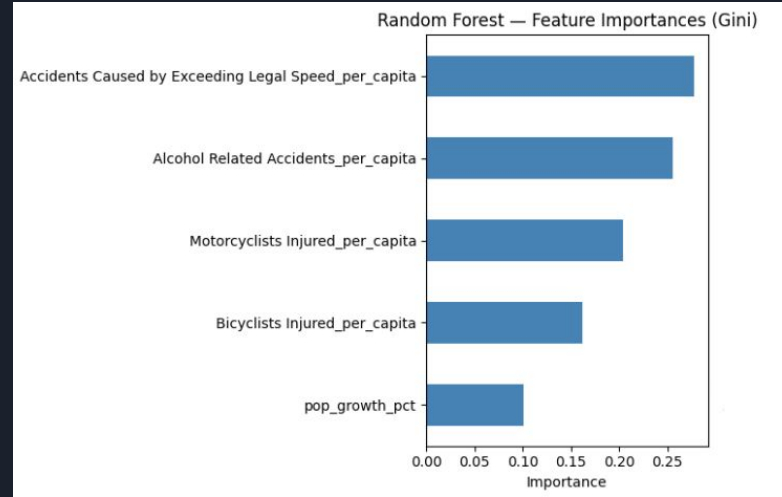
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DETAILED STATISTICAL SUMMARY: Linear Regression
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                        OLS Regression Results
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Dep. Variable:      Traffic Accidents    R-squared:                0.872
Model:              OLS                  Adj. R-squared:           0.836
Method:              Least Squares        F-statistic:              24.35
Date:                Thu, 30 Apr 2026     Prob (F-statistic):       1.15e-09
Time:                19:05:27             Log-Likelihood:           -288.01
No. Observations:    33                  AIC:                      592.0
Df Residuals:        25                  BIC:                      604.0
Df Model:             7
Covariance Type:     nonrobust
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                        coef      std err      t      P>|t|      [0.025      0.975]
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const                8.914e+05    3.51e+05    2.540    0.018    1.68e+05    1.61e+06
log_population        -6.882e+04    2.76e+04   -2.494    0.020   -1.26e+05   -1.2e+04
population_change      0.5202      0.237      2.199    0.037      0.033      1.007
population_growth_pct  -3423.4693   1495.111   -2.290    0.031   -6502.707   -344.231
traffic_accidents_lag1  0.7575      0.179      4.232    0.000      0.389      1.126
traffic_accidents_growth_lag1  39.4888    41.613     0.949    0.352     -46.215    125.193
year_trend            2320.5817    827.551     2.804    0.010     616.209    4024.955
...
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 8.27e+08. This might indicate that there are
strong multicollinearity or other numerical problems.
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Model 2

Our goal when we started making the classification model was to develop a binary classification model to predict high versus low traffic fatality risk years in Mecklenburg County, North Carolina.

The first thing we did was set the target, classifying years as 'high risk' or 'low risk'. Years were labeled using the median threshold of Traffic Accident Fatalities across all years. All models were assessed using 5-fold Stratified Cross-Validation, and performance was measured across three metrics: Accuracy, F1 Score, and ROC-AUC. Logistic ended up being the least accurate, with RF and KNN being similar. We selected RF due to its interpretability.



Model Comparisons

Model	Accuracy	F1	ROC-AUC
LR	0.608	0.604	0.615
RF	0.678	0.672	0.675
KNN	0.697	0.617	0.700



Conclusion

Despite the limitations, we managed to make a Linear Regression model that predicted decently given the data. The model's predictions deviated from actual crash counts by approximately 9,898 crashes on average. Given that Mecklenburg County reports tens of thousands of crashes annually, this error margin represents a reasonable fit. It seems to have a decent prediction of the overall trend but is not fully accurate to the volatility of change in crashes year by year. The model worked but could use more time and data in the future to allow for better fine-tuning and accuracy.

The classification model provided its own challenges in creation leading to us dropping some data as it was a direct match to the target variable leading to a faulty model at first. Upon refining the model we were able to find importance values for each of the features we used. Our main conclusion that was drawn was that the police correctly enforcing speeding and alcohol incidents combined with safety innovation will cause risk years to continue to drop. The model was successful, but things like adding the missing pedestrian incident data or adding the populations of other counties could help us make it more accurate.